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PIXEL SENSOR ARRAYS AND METHODS OF FORMATION

Abstract

An image sensor device may include one or more types of antireflection structures on a metal grid surrounding the pixel sensors in a pixel sensor array of the image sensor device. The antireflection structures may include an antireflective layer, nanostructures extending from the antireflective layer, and/or a plurality of cavities formed in the metal grid structure. The antireflection structures may be included in and/or on one or more surfaces of the metal grid structure to reduce the reflection of incident light, which may reduce the likelihood and/or magnitude of optical crosstalk between adjacent pixel sensors in the pixel sensor array.

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Background/Summary

BACKGROUND

[0001] A complementary metal oxide semiconductor (CMOS) image sensor may include a plurality of pixel sensors. A pixel sensor of the CMOS image sensor may include a transfer gate transistor, which may include a photodiode configured to convert photons of incident light into a photocurrent of electrons and a transfer gate configured to control the flow of the photocurrent between the photodiode and a drain region. The drain region may be configured to receive the photocurrent such that the photocurrent can be measured and/or transferred to other areas of the CMOS image sensor.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0002] Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It should be noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

[0003] FIG. 1 is a diagram of an example of a pixel sensor described herein.

[0004] FIGS. 2A-2C are diagrams of examples of an image sensor device described herein.

[0005] FIGS. 3A-3D are diagrams of examples of pixel sensor arrays of a sensor die described herein.

[0006] FIGS. 4A-4D are diagrams of examples of pixel sensor arrays of a sensor die described herein.

[0007] FIGS. 5A-5J are diagrams of examples of pixel sensor arrays of a sensor die described herein.

[0008] FIGS. 6A-6E are diagrams of an example implementation of forming a circuitry die (or a portion thereof) described herein.

[0009] FIGS. 7A-7F are diagrams of an example implementation of forming a sensor die (or a portion thereof) described herein.

[0010] FIGS. 8A-8F are diagrams of an example implementation of forming an image sensor device (or a portion thereof) described herein.

[0011] FIGS. 9A-9F are diagrams of an example implementation of forming nanostructures on a metal grid structure of a pixel sensor array described herein.

[0012] FIGS. 10A-10D are diagrams of an example implementation of forming nanostructures on a metal grid structure of a pixel sensor array described herein.

[0013] FIGS. 11A-11E are diagrams of an example implementation of forming antireflective layers on a metal grid structure of a pixel sensor array described herein.

[0014] FIGS. 12A-12C are diagrams of an example implementation of forming antireflective layers on a metal grid structure of a pixel sensor array described herein.

[0015] FIGS. 13A-13D are diagrams of an example implementation of forming cavities in a metal grid structure of a pixel sensor array described herein.

[0016] FIGS. 14A and 14B are diagrams of an example implementation of forming cavities in a metal grid structure of a pixel sensor array described herein.

[0017] FIG. 15 is a flowchart of an example process associated with forming a metal grid structure of a pixel sensor array.

DETAILED DESCRIPTION

[0018] The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature

over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

[0019] Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

[0020] An image sensor device (e.g., a complementary metal oxide semiconductor (CMOS) image sensor device or another type of image sensor device) is a type of electronic semiconductor device that uses pixel sensors to generate a photocurrent based on light received at the pixel sensors. The magnitude of the photocurrent may be based on the intensity of the light, based on the wavelength of the light, and/or based on another attribute of the light. The photocurrent is then processed to generate an electronic image, an electronic video, and/or another type of electronic signal.

[0021] Optical crosstalk can occur between adjacent pixel sensors in a pixel sensor array of an image sensor device. Optical crosstalk is a phenomenon whereby incident light passes through a pixel sensor at a non-orthogonal angle and is at least partially absorbed by a photodiode of an adjacent pixel sensor. Optical crosstalk can degrade the spatial resolution of the image sensor device, can reduce overall sensitivity of the image sensor device, can cause color mixing between pixel sensors of the image sensor device, and/or can lead to image noise after color correction.

[0022] In some cases, one or more structures are included in a pixel sensor array to reduce and/or minimize optical crosstalk between pixel sensors of the pixel sensor array. For example, a deep trench isolation (DTI) structure may be formed in a substrate such that the DTI structure surrounds the photodiodes of the pixel sensors in the pixel sensor array. As another example, a metal grid structure may be formed above the substrate, and may surround the pixel sensors in a similar manner as the DTI structure. The metal grid structure may be formed of one or more metal materials to facilitate reflection of incident light toward the photodiodes of the pixel sensors, thereby improving the quantum efficiency (QE) of the pixel array. However, some incident light may be reflected away from a pixel sensor and toward another pixel sensor, resulting in the incident light passing through multiple different types of color filters. This may result in optical crosstalk and associated color mixing between pixel sensors.

[0023] As described herein, an image sensor device may include one or more types of antireflection structures on a metal grid surrounding the pixel sensors in a pixel sensor array of the image sensor device. The antireflection structures may include an antireflective layer, nanostructures extending from the antireflective layer, and/or a plurality of cavities formed in the metal grid structure. The antireflection structures may be included in and/or on one or more surfaces of the metal grid structure to reduce the reflection of incident light, which may reduce the likelihood and/or magnitude of optical crosstalk between adjacent pixel sensors in the pixel sensor array.

[0024] FIG. 1 is a diagram of an example of a pixel sensor **100** described herein. The pixel sensor **100** may include a front side pixel sensor (e.g., a pixel sensor that is configured to receive photons of light from a front side of a sensor die), a back side pixel sensor (e.g., a pixel sensor that is configured to receive photons of light from a back side of a sensor die), and/or another type of pixel sensor. The pixel sensor **100** may be electrically connected to a supply voltage (V_{sub}.dd) **102** and an electrical ground **104**.

[0025] The pixel sensor **100** includes a sensing region **106** that may be configured to sense and/or accumulate incident light (e.g., light directed toward the pixel sensor **100**). The pixel sensor **100** also includes a control circuitry region **108**. The control circuitry region **108** is electrically connected with the sensing region **106** and is configured to receive a photocurrent **110** that is generated by the sensing region **106**. Moreover, the control circuitry region **108** is configured to transfer the photocurrent **110** from the sensing region **106** to downstream circuits such as amplifiers or analog-to-digital (AD) converters, among other examples.

[0026] The sensing region **106** includes a photodiode **112**. The photodiode **112** may absorb and accumulate photons of the incident light, and may generate the photocurrent **110** based on absorbed photons. The magnitude of the photocurrent **110** is based on the amount of light collected in the photodiode **112**. Thus, the accumulation of photons in the photodiode **112** generates a build-up of electrical charge that represents the intensity or brightness of the incident light (e.g., a greater amount of charge may correspond to a greater intensity or brightness, and a lower amount of charge may correspond to a lower intensity or brightness).

[0027] The photodiode **112** is electrically connected with a source of a transfer gate **114** in the control circuitry region **108**. The transfer gate **114** is configured to control the transfer of the photocurrent **110** from the photodiode **112**. The photocurrent **110** is provided from the source of the transfer gate **114** to a drain of the transfer gate **114** based on selectively switching a gate of the transfer gate **114**. The gate of the transfer gate **114** may be selectively switched by applying a transfer voltage ($V_{sub,tx}$) **116** to the transfer gate **114**. In some implementations, the transfer voltage **116** being applied to the transfer gate **114** causes a conductive channel to form between the source and the drain of the transfer gate **114**, which enables the photocurrent **110** to traverse along the conductive channel from the source to the drain. In some implementations, the transfer voltage **116** being removed from the transfer gate **114** (or the absence of the transfer voltage **116**) causes the conductive channel to be removed such that the photocurrent **110** cannot pass from the source to the drain.

[0028] The control circuitry region **108** further includes a reset gate **118**. The reset gate **118** is electrically connected to the supply voltage **102**. The reset gate **118** may be controlled by a reset voltage ($V_{sub,rst}$) **120**. The transfer gate **114** and the reset gate **118** may be electrically coupled with a floating diffusion node **122**. The reset voltage **120** may be applied to the reset gate **118** to pull the drain of the transfer gate **114** to a high voltage (e.g., to the supply voltage **102**) to “reset” the floating diffusion node **122** (e.g., by draining any residual charge in the floating diffusion node **122**) prior to activation of the transfer gate **114** to transfer the photocurrent **110** from the photodiode **112** to the floating diffusion node **122**.

[0029] The photocurrent **110** may be used to apply a floating diffusion voltage ($V_{sub,fd}$) to a source follower gate **124** of the control circuitry region **108**. This permits the photocurrent **110** to be observed without removing or discharging the photocurrent **110** from the floating diffusion node **122**. The reset gate **118** may instead be used to remove or discharge the photocurrent **110** from the floating diffusion node **122**.

[0030] The source follower gate **124** functions as a high impedance amplifier for the pixel sensor **100**. The source follower gate **124** provides a voltage to current conversion of the floating diffusion voltage. The output of the source follower gate **124** is electrically connected with a row select gate **126**, which is configured to control the flow of the photocurrent **110** to external circuitry. The row select gate **126** is controlled by selectively applying a select voltage ($V_{sub,di}$) **128** to the gate of the row select gate **126**. This permits the photocurrent **110** to flow to an output **130** of the pixel sensor **100**.

[0031] As indicated above, FIG. 1 is provided as an example. Other examples may differ from what is described with regard to FIG. 1.

[0032] FIGS. 2A-2C are diagrams of examples **200** of an image sensor device described herein. As shown in FIG. 2A, an image sensor device may be formed by bonding a circuitry wafer **202** and a

sensor wafer **204**. For example, a bonding tool may be used to perform a bonding operation to bond the circuitry wafer **202** and the sensor wafer **204** using a metal-to-metal bonding technique, a dielectric-to-dielectric bonding technique, and/or another bonding technique. In the bonding operation, circuitry dies **206** on the circuitry wafer **202** are bonded with associated sensor dies **208** on the sensor wafer **204** to image sensor devices **210**. The image sensor devices **210** are then diced and packaged. Other processing steps may be performed to form the image sensor devices **210**. [0033] Each image sensor device **210** includes a circuitry die **206** and a sensor die **208**. The circuitry die **206** and the sensor die **208** may be stacked or vertically arranged in the image sensor device **210**. The sensor die **208** includes a pixel sensor array that includes a plurality of pixel sensors **100**, or portions of a plurality of pixel sensors **100**. In particular, the pixel sensor array includes at least the sensing regions **106** (and thus, the photodiodes **112**) of the pixel sensors **100**. Accordingly, the sensor die **208** primarily is configured to sense photons of incident light and convert the photons to a photocurrent **110**.

[0034] The circuitry die **206** includes circuitry that is configured to measure, manipulate, and/or otherwise use the photocurrent **110**. Moreover, the circuitry die **206** includes at least a subset of the transistors of the control circuitry regions **108** of the pixel sensors **100**. For example, the circuitry die **206** may include the row select gates **126** of the pixel sensors **100**, the source follower gates **124** of the pixel sensor, and/or a combination thereof. This provides increased area on the sensor die **208** for the photodiodes **112**, which enables the size of the photodiodes **112** to be increased to increase the sensitivity and/or overall performance of the light sensing performance of the pixel sensor, and/or enables the size of the pixel sensors **100** to be decreased while maintaining the same size for the photodiodes **112**.

[0035] As further shown in FIG. 2A, the circuitry die **206** may include a device layer **212** and an interconnect layer **214**. The device layer **212** may include the devices (e.g., transistors) of the circuitry die **206**, and the interconnect layer **214** may include interconnects that enable signals and/or power to be provided to and/or from the devices in the device layer **212**. The sensor die **208** may also include a device layer **216** and an interconnect layer **218**. The device layer **216** may include portions of the pixel sensors **100**, including the photodiodes **112**, the transfer gates **114**, and the floating diffusion nodes **122**, among other examples. The interconnect layer **218** may include interconnects that enable signals and/or power to be provided to and/or from the device layer **216**.

[0036] The circuitry die **206** and the sensor die **208** may be bonded at a bonding interface **220**, which may be included between the interconnect layers **214** and **218**, and/or may be included in a portion of the interconnect layers **214** and/or **218**. The bonding interface **220** may include bonding pads, bonding vias, bonding dielectric layers, and/or other bonding structures.

[0037] FIG. 2B is a top-down view of an example pixel sensor array **222** included on a sensor die **208**. The pixel sensor array **222** may be included on a sensor die **208** of an image sensor device **210**. As shown in FIG. 2B, the pixel sensor array **222** may include a plurality of pixel sensors **100** (or portions of the plurality of plurality of pixel sensors **100**). For example, the pixel sensor array **222** may include the photodiodes **112** of the pixel sensors **100**. As further shown in FIG. 2B, the pixel sensors **100** may be arranged in a grid. In some implementations, the pixel sensors **100** are square-shaped (as shown in the example in FIG. 2B). In some implementations, the pixel sensors **100** include other shapes such as rectangle shapes, circle shapes, octagon shapes, diamond shapes, and/or other shapes.

[0038] In some implementations, the size of the pixel sensors **100** (e.g., the width or the diameter) of the pixel sensors **100** is approximately 1 micron. In some implementations, the size of the pixel sensors **100** (e.g., the width or the diameter) of the pixel sensors **100** is less than approximately 1 micron. For example, a width of one or more of the pixel sensors **100** may be included in a range of approximately 0.6 microns to approximately 0.7 microns. In these examples, the pixel sensors **100** may be referred to as sub-micron pixel sensors. Sub-micron pixel sensors may decrease the pixel sensor pitch (e.g., the distance between adjacent pixel sensors) in the pixel sensor array **222**, which

may enable increased pixel sensor density in the pixel sensor array **222** (which can increase the performance of the pixel sensor array **222**). However, other values for the range of the size of the pixel sensors **100** are within the scope of the present disclosure.

[0039] Each pixel sensor **100** may be configured to sense a particular wavelength range of incident light associated with a particular color component of the incident light. For example, a pixel sensor **100** may be configured to sense a wavelength range associated with a red component of incident light, and may therefore be referred to as a red pixel sensor. As another example, a pixel sensor **100** may be configured to sense a wavelength range associated with a blue component of incident light, and may therefore be referred to as a blue pixel sensor. As another example, a pixel sensor **100** may be configured to sense a wavelength range associated with a green component of incident light, and may therefore be referred to as a green pixel sensor. In some implementations, a plurality of pixel sensors **100** are configured to sense a wavelength range associated with a near infrared (NIR) component of incident light, and may therefore be referred to as NIR pixel sensors. The NIR pixel sensors may be included in the pixel sensor array **222** to improve low-light performance of the image sensor device **210** and/or to enable night-vision functionality to be realized for the image sensor device **210**.

[0040] As further shown in FIG. 2B, the photodiodes **112** of the pixel sensors **100** may be electrically and optically isolated by a metal grid structure **224** included in the pixel sensor array **222**. The photodiodes **112** may be formed in a substrate of the sensor die **208**, and the metal grid structure **224** may be included above the substrate. The metal grid structure **224** includes a plurality of intersecting metal lines around the perimeters of the pixel sensors **100**. The metal grid structure **224** may be formed of tungsten (W) and/or another suitable metal or metal alloy. The metal grid structure **224** may be included in the pixel sensor array **222** to reduce optical cross-talk between the pixel sensors **100**, which reduces color mixing between the pixel sensors **100**.

[0041] FIG. 2C illustrates a cross-section view of an image sensor device **210**. As shown in FIG. 2C, a circuitry die **206** and a sensor die **208** may be bonded at a bonding interface **220** such that the circuitry die **206** and the sensor die **208** are stacked or vertically arranged in a z-direction in the image sensor device **210**. As further shown in FIG. 2C, the image sensor device **210** includes the pixel sensor array **222** (e.g., including the pixel sensors **100**), a black level region (BLC) region **226** adjacent to (e.g., horizontally adjacent to) the pixel sensor array **222**, a bonding pad region **228** adjacent to (e.g., horizontally adjacent to) the BLC region **226**, and a seal ring region **230** adjacent to (e.g., horizontally adjacent to) the bonding pad region **228**, among other examples.

[0042] As further shown in FIG. 2C, the image sensor device **210** includes a plurality of layers, such as the device layer **212** and the interconnect layer **214** of the circuitry die **206**, and the device layer **216** and the interconnect layer **218** of the sensor die **208**. The device layer **212** of the circuitry die **206** includes a substrate **232** and a dielectric layer **234** above the substrate **232**. The substrate **232** may include silicon (Si) (e.g., a silicon substrate), a material including silicon, a III-V compound semiconductor material such as gallium arsenide (GaAs), a silicon on insulator (SOI), or another type of semiconductor material. The substrate **232** may include a semiconductor layer such as a silicon layer. The dielectric layer **234** may include one or more dielectric materials, such as a silicon oxide (SiOx), a silicon nitride (Si.sub.xN.sub.y), a silicon oxynitride (SiON), tetraethyl orthosilicate oxide, phosphosilicate glass (PSG), borophosphosilicate glass (BPSG), fluorinated silica glass (FSG), and/or carbon doped silicon oxide, among other examples.

[0043] Devices **236** may be included in and/or on the substrate **232** of the device layer **212**. The devices **236** may include one or more application-specific integrated circuit (ASIC) devices, one or more system-on-chip (SOC) devices, one or more transistors, and/or one or more other components configured to measure the magnitude of a photocurrent **110** generated by the pixel sensors **100** to determine light intensity of incident light and/or to generate images and/or video (e.g., digital images, digital video).

[0044] The interconnect layer **214** of the circuitry die **206** may include a dielectric layer **238**, a

bonding layer **240**, a plurality of interconnect structures **242** in the dielectric layer **238**, and a plurality of bonding structures **244** in the bonding layer **240**. The dielectric layer **238** may include one or more interlayer dielectric (ILD) layers, one or more intermetal dielectric (IMD) layers, and/or one or more etch stop layers (ESLs), among other examples. The dielectric layer **238** and the bonding layer **240** may each include one or more dielectric materials, such as a silicon oxide (SiO_x), a silicon nitride (Si₃N₄), a silicon oxynitride (SiON), tetraethyl orthosilicate oxide, phosphosilicate glass (PSG), borophosphosilicate glass (BPSG), fluorinated silica glass (FSG), and/or carbon doped silicon oxide, among other examples.

[0045] The interconnect structures **242** may each include conductive lines, trenches, vias, interconnects, metallization layers, and/or other types of electrically conductive structures that electrically connect the devices **236** to one or more other regions of the circuitry die **206** and/or to one or more regions of the sensor die **208**, among other examples. The bonding structures **244** may each include bonding pads, bonding vias, and/or other types of bonding structures. The interconnect structures **242** and the bonding structures **244** may each include one or more electrically conductive materials, such as, an electrically conductive metal, an electrically conductive metal alloy, an electrically conductive ceramic, tungsten (W), cobalt (Co), ruthenium (Ru), titanium (Ti), aluminum (Al), copper (Cu), and/or gold (Au), among other examples of electrically conductive materials.

[0046] The device layer **216** of the sensor die **208** includes a substrate **246** and a dielectric layer **248** below the substrate **246**. The substrate **246** may include silicon (Si) (e.g., a silicon substrate), a silicon layer or another type of semiconductor layer, a material including silicon, a III-V compound semiconductor material such as gallium arsenide (GaAs), an SOI, or another type of semiconductor material. The dielectric layer **248** may include one or more dielectric materials, such as a silicon oxide (SiO_x), a silicon nitride (Si₃N₄), a silicon oxynitride (SiON), tetraethyl orthosilicate oxide, phosphosilicate glass (PSG), borophosphosilicate glass (BPSG), fluorinated silica glass (FSG), and/or carbon doped silicon oxide, among other examples.

[0047] The photodiodes **112** of the pixel sensors **100** are included in the substrate **246** of the sensor die **208**. The photodiodes **112** may each include one or more doped regions of substrate **246**. The substrate **246** may be doped with a plurality of types of ions to form a p-n junction or a PIN junction (e.g., a junction between a p-type portion, an intrinsic (or undoped) type portion, and an n-type portion) corresponding to a photodiode **112**. For example, the substrate **246** may be doped with an n-type dopant to form a first portion (e.g., an n-type portion) of a photodiode **112** and a p-type dopant to form a second portion (e.g., a p-type portion) of the photodiode **112**. A photodiode **112** may be configured to absorb photons of incident light. The absorption of photons causes the photodiode **112** to accumulate a charge (a photocurrent **110**) due to the photoelectric effect. Here, photons bombard the photodiode **112**, which causes emission of electrons of the photodiode **112**. The emission of electrons causes the formation of electron-hole pairs, where the electrons migrate toward the cathode of the photodiode **112** and the holes migrate toward the anode, which produces the photocurrent **110**.

[0048] The photodiodes **112** may be electrically isolated and/or optically isolated from one another by one or more isolation structures in the substrate **246**. Shallow trench isolation (STI) structures **250** extend into the substrate **246** from a bottom side of the substrate **246** (referred to as the front side of the substrate **246**), and a deep trench isolation (DTI) structure **252** extends into the substrate **246** from a top side of the substrate **246** (referred to as the back side of the substrate **246**) over the STI structures **250**. The combination of the STI structures **250** and the DTI structure **252** in the substrate **246** surround the pixel sensors **100** in the substrate **246** and provide the electrically isolation and/or optically isolation for the pixel sensors **100** in the substrate **246**.

[0049] The STI structures **250** may include one or more dielectric materials, such as a silicon oxide (SiO_x), a silicon nitride (Si₃N₄), and/or a silicon oxynitride (SiON), among other examples. The DTI structure **252** may include elongated structures of dielectric material **254** and a

dielectric liner **256** between the dielectric material **254** and the substrate **246**. The DTI structure **252** extends along the sides of the photodiodes **112** and conforms to the top view shape of the metal grid structure **224** illustrated in FIG. **2B**. The dielectric material **254** may also be included on the top side of the substrate **246** as a buffer layer. The dielectric liner **256** may be included on sidewalls and on a bottom surface of the DTI structure **252**, and may be included as an antireflective coating (ARC) and/or to further facilitate electrical and/or optical isolation of the pixel sensors **100**. In some implementations, the dielectric material **254** includes a silicon oxide (SiOx) (e.g., silicon dioxide (SiO₂)), a silicon nitride (Si_xN_y), a silicon carbide (SiC_x), a hafnium oxide (HfO_x), a silicon oxynitride (SiON), tetraethyl orthosilicate oxide, phosphosilicate glass (PSG), borophosphosilicate glass (BPSG), fluorinated silica glass (FSG), carbon doped silicon oxide, and/or another dielectric material. In some implementations, the dielectric liner **256** may include a high-k dielectric material such as a silicon nitride (Si_xN_y), a hafnium oxide (HfO_x), and/or another high-k dielectric material.

[0050] Transfer gates **114** of the pixel sensors **100** are included in the dielectric layer **248** and on the bottom side of the substrate **246**. The transfer gates **114** are electrically connected to the interconnect layer **218**, which enables inputs (e.g., gate voltages) to be provided to the transfer gates **114** to control the flow of photocurrents **110** from the photodiodes **112** to floating diffusion nodes **122** (not shown) of pixel sensors **100**.

[0051] The interconnect layer **218** may include a dielectric layer **258**, a bonding layer **260**, a plurality of interconnect structures **262** in the dielectric layer **258**, and a plurality of bonding structures **264** in the bonding layer **260**. The dielectric layer **258** may include one or more ILD layers, one or more IMD layers, and/or one or more ESLs, among other examples. The dielectric layer **258** and the bonding layer **260** may each include one or more dielectric materials, such as a silicon oxide (SiOx), a silicon nitride (Si_xN_y), a silicon oxynitride (SiON), tetraethyl orthosilicate oxide, phosphosilicate glass (PSG), borophosphosilicate glass (BPSG), fluorinated silica glass (FSG), and/or carbon doped silicon oxide, among other examples.

[0052] The interconnect structures **262** may each include conductive lines, trenches, vias, interconnects, metallization layers, and/or other types of electrically conductive structures that electrically connect the transfer gates **114** to one or more other regions of the sensor die **208** and/or to one or more regions of the circuitry die **206**, among other examples. The bonding structures **264** may each include bonding pads, bonding vias, and/or other types of bonding structures. The interconnect structures **262** and the bonding structures **264** may each include one or more electrically conductive materials, such as, an electrically conductive metal, an electrically conductive metal alloy, an electrically conductive ceramic, tungsten (W), cobalt (Co), ruthenium (Ru), titanium (Ti), aluminum (Al), copper (Cu), and/or gold (Au), among other examples of electrically conductive materials.

[0053] At the bonding interface **220**, the bonding layers **240** and **260** may be bonded together (e.g., in a dielectric-to-dielectric bond), and the bonding structures **244** and **264** may be bonded together (e.g., in a metal-to-metal bond). Signals and/or power may be provided between the circuitry die **206** and the sensor die **208** through the bonding structures **244** and **264**.

[0054] Above the top side of the substrate **246**, a passivation layer **266** may be included on the buffer layer, and the metal grid structure **224** may be included above the passivation layer **266**. The passivation layer **266** may include an oxide material such as a silicon oxide (SiOx). Additionally and/or alternatively, a silicon nitride (SiNx), a silicon carbide (SiC_x), or a mixture thereof, such as a silicon carbon nitride (SiCN), a silicon oxynitride (SiON), or another dielectric material is used for the passivation layer **266**.

[0055] As shown in FIG. **2C**, sections of the metal grid structure **224** may have an approximately trapezoidal cross-section shape or profile. Additionally and/or alternatively, sections of the metal grid structure **224** may have an approximately square-shaped cross-sectional profile. The sections of the metal grid structure **224** may be located over the DTI structure **252** and may be formed

around the perimeter of the photodiodes **112** of the pixel sensors **100**. Openings in the metal grid structure **224** are included above the photodiodes **112** to enable incident light to pass through the metal grid structure **224** and to the photodiodes **112**. The metal grid structure **224** may be formed of a metal material, such as gold (Au), copper (Cu), silver (Ag), cobalt (Co), tungsten (W), titanium (Ti), ruthenium (Ru), a metal alloy (e.g., aluminum copper (AlCu)), and/or a combination thereof, among other examples.

[0056] Color filter regions **268** of the pixel sensors **100** be included in the openings in the metal grid structure **224**. The color filter regions **268** may be included above the photodiodes **112** of the pixel sensors **100**. The color filter regions **268** may be included above the photodiodes **112**. Each color filter region **268** may be configured to filter incident light to allow a particular wavelength of the incident light to pass to a photodiode **112**. For example, a color filter region **268** may filter incident light to allow red light to pass through the color filter region **268** to an associated photodiode **112**. As another example, a color filter region **268** may filter incident light to allow green light to pass through the color filter region **268** to an associated photodiode **112**. As another example, a color filter region **268** may filter incident light to allow blue light to pass through the color filter region **268** to an associated photodiode **112**.

[0057] A blue color filter region **268** may permit the component of incident light near a 450 nanometer wavelength to pass through and may block other wavelengths from passing. A green color filter region **268** may permit the component of incident light near a 550 nanometer wavelength to pass and may block other wavelengths from passing. A red color filter region **268** may permit the component of incident light near a 650 nanometer wavelength to pass and may block other wavelengths from passing. A yellow color filter region **268** may permit the component of incident light near a 580 nanometer wavelength to pass and may block other wavelengths from passing.

[0058] In some implementations, a color filter region **268** may be non-discriminating or non-filtering, which may define a white pixel sensor. A non-discriminating or non-filtering color filter region **268** may include a material that permits all wavelengths of light to pass into the associated photodiode **112** (e.g., for purposes of determining overall brightness to increase light sensitivity for the image sensor). In some implementations, a color filter region **268** may be a an NIR bandpass color filter region **268**, which may define an NIR pixel sensor. An NIR bandpass color filter region **268** may include a material that permits the portion of incident light in an NIR wavelength range to pass to an associated photodiode **112** while blocking visible light from passing.

[0059] Micro-lenses **270** may be included over and/or on the color filter regions **268**. The micro-lenses **270** may include a respective micro-lens for each of the pixel sensors **100**. A micro-lens may be formed to focus incident light toward a photodiode **112** of an associated pixel sensor **100**.

[0060] As further shown in FIG. 2C, a metal layer **272** may be included above the substrate **246** in the BLC region **226** of the substrate **246**. The metal layer **272** may be included as a light-blocking layer to prevent incident light from entering the portion of substrate **246** in the BLC region **226**. The portion of substrate **246** in the BLC region **226** is thus a sensing region that is kept “dark” so that dark current measurements may be performed in the BLC region **226**. A dark current measurement may be performed to measure the amount of charge (dark current) in the substrate **246** that is generated from sources other than incident light (e.g., from thermal energy in the substrate **246**) so that the dark current measurement may be used for black level correction (or black level calibration) for the pixel sensor array **222**.

[0061] As further shown in FIG. 2C, the bonding pad region **228** may include a plurality of dielectric layers **274**, **276**, **278**, **280**, and **282** that electrically isolate a bonding pad structure **284**. The bonding pad structure **284** is electrically coupled and/or physically coupled with one or more of the interconnect structures **262** in the interconnect layer **218** of the sensor die **208**. A bonding pad opening **286** is included above the bonding pad structure **284** to enable an external electrical connection to be formed to the bonding pad structure **284**.

[0062] The plurality of dielectric layers **274**, **276**, **278**, **280**, and **282** may each include one or more dielectric materials, such as a silicon oxide (SiOx), a silicon nitride (Si.sub.xN.sub.y), a silicon oxynitride (SiON), tetraethyl orthosilicate oxide, phosphosilicate glass (PSG), borophosphosilicate glass (BPSG), fluorinated silica glass (FSG), and/or carbon doped silicon oxide, among other examples. The bonding pad structure **284** may include a metal material, such as gold (Au), copper (Cu), silver (Ag), cobalt (Co), tungsten (W), titanium (Ti), ruthenium (Ru), a metal alloy (e.g., aluminum copper (AlCu)), and/or a combination thereof, among other examples.

[0063] The seal ring region **230** includes a plurality of stacked interconnect structures **242** in the interconnect layer **214** and a plurality of stacked interconnect structures **262** in the interconnect layer **218** to seal the structures and layers of the image sensor device **210** to prevent ingress of humidity and other contaminants, as well as to provide structural rigidity to the image sensor device **210**.

[0064] As indicated above, FIGS. 2A-2C are provided as examples. Other examples may differ from what is described with regard to FIGS. 2A-2C.

[0065] FIGS. 3A-3D are diagrams of examples of pixel sensor arrays **222** of a sensor die **208** described herein. FIGS. 3A-3D illustrate cross-section views of the examples of the pixel sensor arrays **222**. The examples illustrated in FIGS. 3A-3D each include an example implementation of an antireflection structure for the metal grid structure **224** of the pixel sensor arrays **222**. Each example implementation of the antireflection structures described herein may be included in an associated pixel sensor array **222** to reduce, minimize, and/or prevent incident light from being reflected away from a pixel sensor **100** toward another pixel sensor **100**. Thus, each example implementation of the antireflection structures described herein may reduce, minimize, and/or prevent optical crosstalk between the pixel sensors **100** of the associated pixel sensor array **222**. One or more of the examples of the pixel sensor arrays **222** of FIGS. 3A-3D may be included in a sensor die **208** of an image sensor device **210** illustrated and described in connection with FIGS. 2A-2C.

[0066] As shown in FIG. 3A, an example **300** of a pixel sensor array **222** includes a plurality of pixel sensors **100**. The pixel sensors **100** each include a photodiode **112** in the substrate **246** of the sensor die **208**. A DTI structure **252** is included around the photodiodes **112** in the substrate **246**, and a metal grid structure **224** is included on the substrate **246** above the DTI structure **252**. The metal grid structure **224** extends above the substrate **246** and surrounds the photodiodes **112**. Color filter regions **268** are included between the openings of the metal grid structure **224**, and micro-lenses **270** are included on the color filter regions. A passivation layer **302** may additionally be included on the metal grid structure **224**, or may be omitted.

[0067] As further shown in the example **300**, nanostructures **304** are included on top surfaces **306** of the metal grid structure **224**. The nanostructures **304** may include nanotubes, nanowires, and/or another type of nanostructures that extend from the top surfaces **306** of the metal grid structure **224**. The nanostructures **304** may be formed of carbon (C) and/or another suitable material. The nanostructures **304** may reduce optical crosstalk in the pixel sensor array **222** in that the nanostructures **304** may absorb photons of incident light that might otherwise reflect off of the top surfaces **306** of the metal grid structure **224**.

[0068] A length of the nanostructures **304** (indicated in FIG. 3A as dimension D1) may be included in a range of approximately 80 nanometers to approximately 1000 nanometers. If the length of the nanostructures **304** is less than approximately 80 nanometers, the nanostructures **304** may not provide sufficient optical absorption to reduce or prevent optical crosstalk, whereas the nanostructures **304** may reduce absorption of photons in the photodiodes **112** (which may reduce the quantum efficiency of the pixel sensors **100**) if the length of the nanostructures **304** is greater than approximately 1000 nanometers. If the length of the nanostructures **304** is included in the range of approximately 80 nanometers to approximately 1000 nanometers, the nanostructures **304** may provide sufficient optical absorption to reduce or prevent optical crosstalk while enabling a

high quantum efficiency to be achieved for the pixel sensors **100**. However, other values for the length of the nanostructures **304**, and ranges other than approximately 80 nanometers to approximately 1000 nanometers, are within the scope of the present disclosure.

[0069] The nanostructures **304** may extend in a direction that is approximately perpendicular to the top surfaces **306** of the metal grid structure **224**. Additionally and/or alternatively, at least a subset of the nanostructures **304** may extend in a direction that is angled (e.g., angled at approximately 60 degrees, angled at approximately 75 degrees) relative to the top surfaces **306** of the metal grid structure **224**.

[0070] As further shown in FIG. 3A, the metal grid structures **224** in the example **300** of the pixel sensor array **222** may have an approximately trapezoidal cross-section shape or profile. For example, a cross-sectional width of the top surface **306** of the metal grid structure **224** may be less than a cross-sectional width of a bottom surface of the metal grid structure **224**. Thus, the cross-sectional width increases from the top surface **306** to the bottom surface for the metal grid structure **224**.

[0071] FIG. 3B illustrates an example **308** of a pixel sensor array **222**, which is similar to the example **300** of the pixel sensor array **222** except that the example **308** of the pixel sensor array **222** further includes nanostructures **304** on sidewalls **310** of the metal grid structure **224**. Thus, the nanostructures **304** are included on the top surfaces **306** and on the sidewalls **310** of the metal grid structure **224**. The nanostructures **304** may provide further optical crosstalk reduction by absorbing photons of incident light that are directed toward the sidewalls **310** of the metal grid structure **224**.

[0072] FIG. 3C illustrates an example **312** of a pixel sensor array **222**, which is similar to the example **300** of the pixel sensor array **222** except that the metal grid structure **224** in the example **312** of the pixel sensor array **222** has a square-shaped cross-section shape or profile. Thus, the sidewalls **310** of the metal grid structure **224** are approximately perpendicular to the top surfaces **306** of the metal grid structure **224**.

[0073] FIG. 3D illustrates an example **314** of a pixel sensor array **222**, which is similar to the example **312** of the pixel sensor array **222** except that the example **314** of the pixel sensor array **222** further includes nanostructures **304** on the sidewalls **310** of the metal grid structure **224**. Thus, the nanostructures **304** are included on the top surfaces **306** and on the sidewalls **310** of the metal grid structure **224**, where the sidewalls **310** are approximately perpendicular to the top surfaces **306**. The nanostructures **304** may provide further optical crosstalk reduction by absorbing photons of incident light that are directed toward the sidewalls **310** of the metal grid structure **224**.

[0074] As indicated above, FIGS. 3A-3D is provided as examples. Other examples may differ from what is described with regard to FIGS. 3A-3D.

[0075] FIGS. 4A-4D are diagrams of examples of pixel sensor arrays **222** of a sensor die **208** described herein. FIGS. 4A-4D illustrate cross-section views of the examples of the pixel sensor arrays **222**. The examples illustrated in FIGS. 4A-4D each include an example implementation of an antireflection structure for the metal grid structure **224** of the pixel sensor arrays **222**. Each example implementation of the antireflection structures described herein may be included in an associated pixel sensor array **222** to reduce, minimize, and/or prevent incident light from being reflected away from a pixel sensor **100** toward another pixel sensor **100**. Thus, each example implementation of the antireflection structures described herein may reduce, minimize, and/or prevent optical crosstalk between the pixel sensors **100** of the associated pixel sensor array **222**. One or more of the examples of the pixel sensor arrays **222** of FIGS. 4A-4D may be included in a sensor die **208** of an image sensor device **210** illustrated and described in connection with FIGS. 2A-2C.

[0076] As shown in FIG. 4A, an example **400** of a pixel sensor array **222** includes a plurality of pixel sensors **100**. The pixel sensors **100** each include a photodiode **112** in the substrate **246** of the sensor die **208**. A DTI structure **252** is included around the photodiodes **112** in the substrate **246**, and a metal grid structure **224** is included on the substrate **246** above the DTI structure **252**. The

metal grid structure **224** extends above the substrate **246** and surrounds the photodiodes **112**. Color filter regions **268** are included between the openings of the metal grid structure **224**, and microlenses **270** are included on the color filter regions. A passivation layer **302** may additionally be included on the metal grid structure **224**, or may be omitted.

[0077] As further shown in the example **400**, antireflective layers **402** are included on the top surfaces **306** of the metal grid structure **224**. The antireflective layers **402** may each include a layer of antireflective material that absorbs photons of incident light. The layer of antireflective material may have one or more properties that, when included on the metal grid structure **224**, result in cancelation of a broad range of wavelengths of incident light. Examples of such properties include refractive index, optical band gap, and/or thickness, among other examples. The properties of the layer of antireflective material may result in formation of destructive waves that are opposing and out of phase of the incident light, and these destructive waves cancel the incident light, which prevents or reduces the incident light from being reflected off of the metal grid structure **224**. Thus, the antireflective material may be selected based on the material used for the metal grid structure **224**. For example, the antireflective material may be selected such that the refractive index; optical band gap; and/or another property of the antireflective material, in combination with the refractive index; optical band gap; and/or another property of the metal material of the metal grid structure, results in formation of destructive waves that cancel the incident light. Examples of such antireflective materials include zinc oxide (ZnO), aluminum-doped zinc oxide (Al: ZnO or AZO), and/or aluminum oxide (Al.sub.xO.sub.y such as Al.sub.3O.sub.4), among other examples.

[0078] A thickness of the antireflective layers **402** (indicated in FIG. 4A as dimension D2) may be included in a range of approximately 50 nanometers to approximately 200 nanometers. If the thickness of the antireflective layers **402** is less than approximately 50 nanometers or greater than approximately 200 nanometers, the properties of the antireflective layers **402** may not combine with the properties of the metal material of the metal grid structure **224** to cancel out incident light to reduce or prevent optical crosstalk. If the thickness of the antireflective layers **402** is included in the range of approximately 50 nanometers to approximately 200 nanometers, the properties of the antireflective layers **402** may combine with the properties of the metal material of the metal grid structure **224** to cancel out incident light to reduce or prevent optical crosstalk. However, other values for the thickness of the antireflective layers **402**, and ranges other than approximately 50 nanometers to approximately 200 nanometers, are within the scope of the present disclosure.

[0079] As further shown in FIG. 4A, the metal grid structures **224** in the example **400** of the pixel sensor array **222** may have an approximately trapezoidal cross-section shape or profile. For example, a cross-sectional width of the top surface **306** of the metal grid structure **224** may be less than a cross-sectional width of a bottom surface of the metal grid structure **224**. Thus, the cross-sectional width increases from the top surface **306** to the bottom surface for the metal grid structure **224**.

[0080] FIG. 4B illustrates an example **404** of a pixel sensor array **222**, which is similar to the example **400** of the pixel sensor array **222** except that the antireflective layers **402** are included on the sidewalls **310** of the metal grid structure **224** in the example **404** of the pixel sensor array **222**. Thus, the antireflective layers **402** are included on the top surfaces **306** and on the sidewalls **310** of the metal grid structure **224**. The antireflective layers **402** may provide further optical crosstalk reduction by canceling out reflections of incident light that might otherwise reflect off of the sidewalls **310** of the metal grid structure **224**.

[0081] FIG. 4C illustrates an example **406** of a pixel sensor array **222**, which is similar to the example **400** of the pixel sensor array **222** except that the metal grid structure **224** in the example **406** of the pixel sensor array **222** has a square-shaped cross-section shape or profile. Thus, the sidewalls **310** of the metal grid structure **224** are approximately perpendicular to the top surfaces **306** of the metal grid structure **224**.

[0082] FIG. 4D illustrates an example **408** of a pixel sensor array **222**, which is similar to the

example **406** of the pixel sensor array **222** except that the antireflective layers **402** are included on the sidewalls **310** of the metal grid structure **224** in the example **404** of the pixel sensor array **222**. Thus, the antireflective layers **402** are included on the top surfaces **306** and on the sidewalls **310** of the metal grid structure **224**, where the sidewalls **310** are approximately perpendicular to the top surfaces **306**. The antireflective layers **402** may provide further optical crosstalk reduction by canceling out reflections of incident light that might otherwise reflect off of the sidewalls **310** of the metal grid structure **224**.

[0083] As indicated above, FIGS. **4A-4D** is provided as examples. Other examples may differ from what is described with regard to FIGS. **4A-4D**.

[0084] FIGS. **5A-5J** are diagrams of examples of pixel sensor arrays **222** of a sensor die **208** described herein. FIGS. **5A-5J** illustrate cross-section views of the examples of the pixel sensor arrays **222**. The examples illustrated in FIGS. **5A-5J** each include an example implementation of an antireflection structure for the metal grid structure **224** of the pixel sensor arrays **222**. Each example implementation of the antireflection structures described herein may be included in an associated pixel sensor array **222** to reduce, minimize, and/or prevent incident light from being reflected away from a pixel sensor **100** toward another pixel sensor **100**. Thus, each example implementation of the antireflection structures described herein may reduce, minimize, and/or prevent optical crosstalk between the pixel sensors **100** of the associated pixel sensor array **222**. One or more of the examples of the pixel sensor arrays **222** of FIGS. **5A-5J** may be included in a sensor die **208** of an image sensor device **210** illustrated and described in connection with FIGS. **2A-2C**.

[0085] As shown in FIG. **5A**, an example **500** of a pixel sensor array **222** includes a plurality of pixel sensors **100**. The pixel sensors **100** each include a photodiode **112** in the substrate **246** of the sensor die **208**. A DTI structure **252** is included around the photodiodes **112** in the substrate **246**, and a metal grid structure **224** is included on the substrate **246** above the DTI structure **252**. The metal grid structure **224** extends above the substrate **246** and surrounds the photodiodes **112**. Color filter regions **268** are included between the openings of the metal grid structure **224**, and micro-lenses **270** are included on the color filter regions. A passivation layer **302** may additionally be included on the metal grid structure **224**, or may be omitted.

[0086] As further shown in the example **500**, cavities **502** are included in the top surfaces **306** of the metal grid structure **224** as the antireflection structures. The cavities **502** result in the top surfaces **306** of the metal grid structure **224** having a plurality of concave surfaces (or a plurality of recessed surfaces). The cavities **502** may be spaced apart from each other in the top surfaces **306** and may each have an approximately curved, rounded, and/or semi-circular cross-sectional profile. The curved or rounded surfaces of the cavities **502** scatter photons of incident light. Scattering the photons of incident light around a photodiode **112** causes the incident light to be directed in multiple directions off of the top surfaces **306** of the metal grid structure **224**, thereby reducing the concentration of photons reflected toward other photodiodes **112**, and thereby reducing optical crosstalk in the pixel sensor array **222**.

[0087] The cavities **502** may have one or more dimensions, such as a depth (illustrated in FIG. **5A** as dimension **D3**), a spacing between adjacent cavities **502** (illustrated in FIG. **5A** as dimension **D4**), and/or a width (illustrated in FIG. **5A** as dimension **D5**), among other examples. One or more dimensions of the cavities **502**, the quantity of cavities **502**, and/or another parameter of the cavities **502** may be selected to reduce, minimize, and/or prevent optical crosstalk from resulting from reflections of incident light off of the metal grid structure **224**. The one or more dimensions of the cavities **502**, the quantity of cavities **502**, and/or another parameter of the cavities **502** may be selected based on a size of the pixel sensors **100** (e.g., a width of the pixel sensors **100**, based on a wavelength or wavelengths of incident light that is to be sensed by the pixel sensor array **222**, spacing between adjacent pixel sensors **100**, and/or another parameter).

[0088] In some implementations, the depth of a cavity **502** (dimension **D3**) is included in a range of

approximately 1 nanometer to approximately 10 nanometers. However, other values for the range are within the scope of the present disclosure. In some implementations, the spacing between adjacent cavities **502** (dimension **D4**) is included in a range of approximately 1 nanometer to approximately 5 nanometers. However, other values for the range are within the scope of the present disclosure. In some implementations, the width of a cavity **502** (dimension **D5**) is included in a range of approximately 1 nanometer to approximately 50 nanometers. However, other values for the range are within the scope of the present disclosure.

[0089] As further shown in FIG. 5A, the metal grid structures **224** in the example **500** of the pixel sensor array **222** may have an approximately trapezoidal cross-section shape or profile. For example, a cross-sectional width of the top surface **306** of the metal grid structure **224** may be less than a cross-sectional width of a bottom surface of the metal grid structure **224**. Thus, the cross-sectional width increases from the top surface **306** to the bottom surface for the metal grid structure **224**.

[0090] FIG. 5B illustrates an example **504** of a pixel sensor array **222**, which is similar to the example **500** of the pixel sensor array **222** except that the cavities **502** have an approximately V-shaped (or triangular) cross-sectional profile in the example **504** of the pixel sensor array **222**. The cavities **502** in the example **504** may have one or more dimensions, such as a depth (illustrated in FIG. 5B as dimension **D6**), a spacing between adjacent cavities **502** (illustrated in FIG. 5B as dimension **D7**), a sidewall angle relative to a bottom surface of the metal grid structure **224** (illustrated in FIG. 5B as dimension **D8**), and/or a width (illustrated in FIG. 5B as dimension **D9**), among other examples. One or more dimensions of the cavities **502**, the quantity of cavities **502**, and/or another parameter of the cavities **502** may be selected to reduce, minimize, and/or prevent optical crosstalk from resulting from reflections of incident light off of the metal grid structure **224**. The one or more dimensions of the cavities **502**, the quantity of cavities **502**, and/or another parameter of the cavities **502** may be selected based on a size of the pixel sensors **100** (e.g., a width of the pixel sensors **100**, based on a wavelength or wavelengths of incident light that is to be sensed by the pixel sensor array **222**, spacing between adjacent pixel sensors **100**, and/or another parameter).

[0091] In some implementations, the depth of a cavity **502** (dimension **D6**) is included in a range of approximately 1 nanometer to approximately 10 nanometers. However, other values for the range are within the scope of the present disclosure. In some implementations, the spacing between adjacent cavities **502** (dimension **D7**) is included in a range of approximately 1 nanometer to approximately 5 nanometers. However, other values for the range are within the scope of the present disclosure. In some implementations, the angle of the sidewalls of a cavity **502** (dimension **D8**) is included in a range of approximately 30 degrees to approximately 75 degrees. However, other values for the range are within the scope of the present disclosure. In some implementations, the width of a cavity **502** (dimension **D9**) is included in a range of approximately 1 nanometer to approximately 50 nanometers. However, other values for the range are within the scope of the present disclosure.

[0092] FIG. 5C illustrates an example **506** of a pixel sensor array **222**, which is similar to the example **500** of the pixel sensor array **222** except that the cavities **502** having approximately curved, rounded, and/or semi-circular cross-sectional profiles are also included in the sidewalls **310** of the metal grid structure **224** in the example **504** of the pixel sensor array **222**. Thus, the cavities **502** having approximately curved, rounded, and/or semi-circular cross-sectional profiles are included in the top surfaces **306** and the sidewalls **310** of the metal grid structure **224**. The cavities **502** in the sidewalls **310** may provide further optical crosstalk reduction by scattering incident light directed to the sidewalls **310** of the metal grid structure **224**.

[0093] FIG. 5D illustrates an example **508** of a pixel sensor array **222**, which is similar to the example **504** of the pixel sensor array **222** except that the cavities **502** having approximately V-shaped (or triangular) cross-sectional profiles are also included in the sidewalls **310** of the metal

grid structure **224** in the example **504** of the pixel sensor array **222**. Thus, the cavities **502** having approximately V-shaped (or triangular) cross-sectional profiles are included in the top surfaces **306** and the sidewalls **310** of the metal grid structure **224**. The cavities **502** in the sidewalls **310** may provide further optical crosstalk reduction by scattering incident light directed to the sidewalls **310** of the metal grid structure **224**.

[0094] FIG. 5E illustrates an example **510** of a pixel sensor array **222**, which is similar to the examples **506** and **508** of the pixel sensor arrays **222** except that the example **510** of the pixel sensor array **222** in FIG. 5E includes cavities **502** in the top surfaces **306** of the metal grid structure **224** and cavities **502** in the sidewalls **310** of the metal grid structure **224** that have different cross-sectional profiles. For example, the cavities **502** in the top surfaces **306** of the metal grid structure **224** may have approximately V-shaped (or triangular) cross-sectional profiles, and the cavities **502** in the sidewalls **310** of the metal grid structure **224** may have approximately curved, rounded, and/or semi-circular cross-sectional profiles (as illustrated in the example in FIG. 5E). As another example, the cavities **502** in the top surfaces **306** of the metal grid structure **224** may have approximately curved, rounded, and/or semi-circular cross-sectional profiles, and the cavities **502** in the sidewalls **310** of the metal grid structure **224** may have approximately V-shaped (or triangular) cross-sectional profiles. This enables greater flexibility in configuring the metal grid structure **224** for further optical crosstalk reduction for the pixel sensor array **222**.

[0095] FIG. 5F illustrates an example **512** of a pixel sensor array **222**, which is similar to the example **500** of the pixel sensor array **222** except that the metal grid structure **224** in the example **512** of the pixel sensor array **222** has a square-shaped cross-section shape or profile. Thus, the sidewalls **310** of the metal grid structure **224** are approximately perpendicular to the top surfaces **306** of the metal grid structure **224**.

[0096] FIG. 5G illustrates an example **514** of a pixel sensor array **222**, which is similar to the example **504** of the pixel sensor array **222** except that the metal grid structure **224** in the example **514** of the pixel sensor array **222** has a square-shaped cross-section shape or profile. Thus, the sidewalls **310** of the metal grid structure **224** are approximately perpendicular to the top surfaces **306** of the metal grid structure **224**.

[0097] FIG. 5H illustrates an example **516** of a pixel sensor array **222**, which is similar to the example **506** of the pixel sensor array **222** except that the metal grid structure **224** in the example **516** of the pixel sensor array **222** has a square-shaped cross-section shape or profile. Thus, the sidewalls **310** of the metal grid structure **224** are approximately perpendicular to the top surfaces **306** of the metal grid structure **224**, and the cavities **502** in the sidewalls **310** of the metal grid structure **224** may have approximately curved, rounded, and/or semi-circular cross-sectional profiles.

[0098] FIG. 5I illustrates an example **518** of a pixel sensor array **222**, which is similar to the example **508** of the pixel sensor array **222** except that the metal grid structure **224** in the example **518** of the pixel sensor array **222** has a square-shaped cross-section shape or profile. Thus, the sidewalls **310** of the metal grid structure **224** are approximately perpendicular to the top surfaces **306** of the metal grid structure **224**, and the cavities **502** in the sidewalls **310** of the metal grid structure **224** may have approximately V-shaped (or triangular) cross-sectional profiles.

[0099] FIG. 5J illustrates an example **520** of a pixel sensor array **222**, which is similar to the example **510** of the pixel sensor array **222** except that the metal grid structure **224** in the example **520** of the pixel sensor array **222** has a square-shaped cross-section shape or profile. Thus, the sidewalls **310** of the metal grid structure **224** are approximately perpendicular to the top surfaces **306** of the metal grid structure **224**. The cavities **502** in the sidewalls **310** may have approximately V-shaped (or triangular) cross-sectional profiles, or may have approximately curved, rounded, and/or semi-circular cross-sectional profiles.

[0100] As indicated above, FIGS. 5A-5J is provided as examples. Other examples may differ from what is described with regard to FIGS. 5A-5J.

[0101] FIGS. 6A-6E are diagrams of an example implementation **600** of forming a circuitry die **206** (or a portion thereof) described herein. In some implementations, one or more of the semiconductor processing operations described in connection with FIGS. 6A-6E may be performed using one or more semiconductor processing tools, such as a deposition tool, an exposure tool, a developer tool, an etch tool, a planarization tool, a plating tool, an ion implantation tool, and/or a wafer/die transport tool, among other examples.

[0102] Turning to FIG. 6A, the substrate **232** of the device layer **212** of the circuitry die **206** is provided. The substrate **232** may be provided in the form of a semiconductor wafer such as a silicon (Si) wafer may be provided as an SOI wafer, and/or another type of semiconductor work piece.

[0103] As shown in FIG. 6B, one or more devices **236** may be formed in and/or on the substrate **232**. One or more semiconductor processing tools may be used to form one or more portions of the devices **236**. For example, a deposition tool may be used to perform various deposition operations to deposit layers of the devices **236**, and/or to deposit photoresist layers for etching the substrate **232** and/or portions of the deposited layers. As another example, an exposure tool may be used to expose the photoresist layers to form patterns in the photoresist layers. As another example, a developer tool may develop the patterns in the photoresist layers. As another example, an etch tool may be used to etch the substrate **232** and/or portions of the deposited layers to form the devices **236**. As another example, a planarization tool may be used to planarize portions of the devices **236**. As another example, a plating tool may be used to deposit metal structures and/or layers of the devices **236**.

[0104] As further shown in FIG. 6B, a dielectric layer **234** may be deposited over and/or on the substrate **232** and over and/or on the devices **236**. A deposition tool may be used to deposit the dielectric layer **234** using a physical vapor deposition (PVD) technique, an atomic layer deposition (ALD) technique, a chemical vapor deposition (CVD) technique, an oxidation technique, another type of deposition technique. In some implementations, a planarization tool may be used to planarize the dielectric layer **234** after the dielectric layer **234** is deposited.

[0105] As shown in FIG. 6C, a first portion of an interconnect layer **214** of the circuitry die **206** is formed above the device layer **212**. To form the first portion of the interconnect layer **214**, a deposition tool may be used to deposit a dielectric layer **238** (which may include one or more ILD layers, one or more IMD layers, one or more ESLs, and/or one or more of another type of dielectric layer) using a PVD technique, an ALD technique, a CVD technique, an oxidation technique, another deposition technique. In some implementations, a planarization tool may be used to planarize the dielectric layer **238** after the dielectric layer **238** is deposited.

[0106] A deposition tool, an exposure tool, a developer tool, an etch tool, a planarization tool, a plating tool, and/or another semiconductor processing tool may be used to perform various operations to form interconnect structures **242** in the first portion of the interconnect layer **214**. A deposition tool and/or a plating tool may be used to deposit the interconnect structures **242** using a PVD technique, an ALD technique, a CVD technique, an electroplating technique, and/or another deposition technique. In some implementations, a planarization tool may be used to planarize the interconnect structures **242** after the interconnect structures **242** are deposited.

[0107] In some implementations, first portion of the interconnect layer **214** is built up in the z-direction in a plurality of via layers (V-layers) and metallization layers (M-layers). For example, a first portion of the dielectric layer **238** may be formed, recesses may be formed in the first portion of the dielectric layer **238**, and first interconnect structures **242** (e.g., a V0 via layer, an M0 metallization layer) may be formed in the recesses. A second portion of the dielectric layer **238** may be formed, recesses may be formed in the second portion of the dielectric layer **238**, and second interconnect structures **242** (e.g., a V1 via layer, an M1 metallization layer) may be formed in the recesses. The remaining via layers and/or metallization layers of the first portion of the interconnect layer **214** may be formed in a similar manner.

[0108] As shown in FIGS. 6D and 6E, a second portion of the interconnect layer **214** may be formed, and the second portion of the interconnect layer **214** may include a bonding layer **240** and bonding structures **244**. As shown in FIG. 6D, the bonding layer **240** may be formed over and/or on the dielectric layer **238**, and over and/or on the top-most interconnect structures **242**. A deposition tool may be used to deposit the bonding layer **240** using a PVD technique, an ALD technique, a CVD technique, an oxidation technique, another deposition technique. In some implementations, a planarization tool may be used to planarize the bonding layer **240** after the bonding layer **240** is deposited.

[0109] As shown in FIG. 6E, the bonding structures **244** may be formed in the bonding layer **240**. For example, a deposition tool, an exposure tool, and a developer tool may be used to form a patterned masking layer on the bonding layer **240**. An etch tool may be used to etch the bonding layer **240** (e.g., using a wet etch technique, a dry etch technique) to form recesses in the bonding layer **240**. A deposition tool and/or a plating tool may be used to deposit the bonding structures **244** in the recess using a CVD technique, a PVD technique, an ALD technique, an electroplating technique, and/or another deposition technique. In some implementations, a planarization tool may perform a planarization operation to planarize the bonding structures **244** after the bonding structures **244** are deposited.

[0110] As indicated above, FIGS. 6A-6E are provided as an example. Other examples may differ from what is described with regard to FIGS. 6A-6E.

[0111] FIGS. 7A-7F are diagrams of an example implementation **700** of forming a sensor die **208** (or a portion thereof) described herein. In some implementations, one or more of the semiconductor processing operations described in connection with FIGS. 7A-7F may be performed using one or more semiconductor processing tools, such as a deposition tool, an exposure tool, a developer tool, an etch tool, a planarization tool, a plating tool, an ion implantation tool, and/or a wafer/die transport tool, among other examples.

[0112] Turning to FIG. 7A, the substrate **246** of the device layer **216** of the sensor die **208** is provided. The substrate **246** may be provided in the form of a semiconductor wafer such as a silicon (Si) wafer may be provided as an SOI wafer, and/or another type of semiconductor work piece.

[0113] As shown in FIG. 7B, photodiodes **112** of pixel sensors **100** of a pixel sensor array **222** of the sensor die **208** may be formed in the substrate **246** in the device layer **216** of the sensor die **208**. In some implementations, an ion implantation tool may be used to implant ions into the substrate **246** to form a P-N junction between a p-doped region of the substrate **246** and an n-doped region of the substrate **246**, or to form a P-I-N junction between p-doped region of the substrate **246**, an n-doped region of the substrate **246**, and an intrinsic (e.g., undoped) semiconductor region for a photodiode **112**.

[0114] As further shown in FIG. 7B, STI structures **250** may be formed in the substrate **246** (e.g., from the front side of the substrate **246**) such that the STI structures **250** are located between the photodiodes **112**. In some implementations, the STI structures **250** are formed after the photodiodes **112** are formed. In some implementations, the STI structures **250** are formed prior to formation of the photodiodes **112**. A deposition tool, an exposure tool, and a developer tool may be used to form a patterned masking layer on the substrate **246**. An etch tool may be used to etch into the substrate **246** from the front side of the substrate **246** (e.g., using a wet etch technique, a dry etch technique) to form the recesses in the front side of the substrate **246**. A deposition tool may be used to deposit the STI structures **250** in the recesses using a CVD technique, a PVD technique, an ALD technique, an oxidation technique, and/or another deposition technique. In some implementations, a planarization tool may perform a planarization operation to planarize the STI structures **250** after the STI structures **250** are deposited.

[0115] As shown in FIG. 7C, transfer gates **114** of the pixel sensors **100** may be formed over and/or on the front side surface of the substrate **246**. Forming a transfer gate **114** may include depositing a

gate dielectric on the front side surface of the substrate **246**, depositing a gate electrode on the gate dielectric layer, and/or forming sidewall spacers on sidewalls of the gate electrode, among other examples.

[0116] As further shown in FIG. 7C, a dielectric layer **248** may be formed over and/or on the front side of the substrate **246**, and over and/or on the transfer gates **114**. A deposition tool may be used to deposit the dielectric layer **248** using a CVD technique, a PVD technique, an ALD technique, an electroplating technique, and/or another deposition technique. In some implementations, a planarization tool may perform a planarization operation to planarize the dielectric layer **248** after the dielectric layer **248** is deposited.

[0117] As shown in FIG. 7D, a first portion of an interconnect layer **218** of the sensor die **208** is formed above the device layer **216**. To form the first portion of the interconnect layer **218**, a deposition tool may be used to deposit a dielectric layer **258** (which may include one or more ILD layers, one or more IMD layers, one or more ESLs, and/or one or more of another type of dielectric layer) using a PVD technique, an ALD technique, a CVD technique, an oxidation technique, another deposition technique. In some implementations, a planarization tool may be used to planarize the dielectric layer **258** after the dielectric layer **258** is deposited.

[0118] A deposition tool, an exposure tool, a developer tool, an etch tool, a planarization tool, a plating tool, and/or another semiconductor processing tool may be used to perform various operations to form interconnect structures **262** in the first portion of the interconnect layer **218**. A deposition tool and/or a plating tool may be used to deposit the interconnect structures **262** using a PVD technique, an ALD technique, a CVD technique, an electroplating technique, and/or another deposition technique. In some implementations, a planarization tool may be used to planarize the interconnect structures **262** after the interconnect structures **262** are deposited.

[0119] In some implementations, first portion of the interconnect layer **218** is built up in the z-direction in a plurality of via layers (V-layers) and metallization layers (M-layers). For example, a first portion of the dielectric layer **258** may be formed, recesses may be formed in the first portion of the dielectric layer **258**, and first interconnect structures **262** (e.g., a V0 via layer, an M0 metallization layer) may be formed in the recesses. A second portion of the dielectric layer **258** may be formed, recesses may be formed in the second portion of the dielectric layer **258**, and second interconnect structures **262** (e.g., a V1 via layer, an M1 metallization layer) may be formed in the recesses. The remaining via layers and/or metallization layers of the first portion of the interconnect layer **218** may be formed in a similar manner.

[0120] As shown in FIGS. 7E and 7F, a second portion of the interconnect layer **218** may be formed, and the second portion of the interconnect layer **218** may include a bonding layer **260** and bonding structures **264**. As shown in FIG. 7E, the bonding layer **260** may be formed over and/or on the dielectric layer **260**, and over and/or on the top-most interconnect structures **262**. A deposition tool may be used to deposit the bonding layer **260** using a PVD technique, an ALD technique, a CVD technique, an oxidation technique, another deposition technique. In some implementations, a planarization tool may be used to planarize the bonding layer **260** after the bonding layer **260** is deposited.

[0121] As shown in FIG. 7F, the bonding structures **264** may be formed in the bonding layer **260**. For example, a deposition tool, an exposure tool, and a developer tool may be used to form a patterned masking layer on the bonding layer **260**. An etch tool may be used to etch the bonding layer **260** (e.g., using a wet etch technique, a dry etch technique) to form recesses in the bonding layer **260**. A deposition tool and/or a plating tool may be used to deposit the bonding structures **264** in the recess using a CVD technique, a PVD technique, an ALD technique, an electroplating technique, and/or another deposition technique. In some implementations, a planarization tool may perform a planarization operation to planarize the bonding structures **264** after the bonding structures **264** are deposited.

[0122] As indicated above, FIGS. 7A-7F are provided as an example. Other examples may differ

from what is described with regard to FIGS. 7A-7F.

[0123] FIGS. 8A-8F are diagrams of an example implementation **800** of forming an image sensor device **210** (or a portion thereof) described herein. In some implementations, one or more of the semiconductor processing operations described in connection with FIGS. 8A-8F may be performed using one or more semiconductor processing tools, such as a deposition tool, an exposure tool, a developer tool, an etch tool, a planarization tool, a plating tool, an ion implantation tool, and/or a wafer/die transport tool, among other examples.

[0124] As shown in FIGS. 8A and 8B, a bonding operation is performed to bond a circuitry die **206** and a sensor die **208** to form the image sensor device **210**. The circuitry die **206** and the sensor die **208** may be bonded at a bonding interface **220**, which may include the bonding layers **240** and **260** (respectively of the circuitry die **206** and the sensor die **208**), and the bonding structures **244** and **264** (respectively of the circuitry die **206** and the sensor die **208**). A bonding tool may be used to form a dielectric-to-dielectric bond between the bonding layers **240** and **260** at the bonding interface **220**, and to form a metal-to-metal bond between the bonding structures **244** and **264** at the bonding interface **220**.

[0125] As shown in FIG. 8B, after bonding, the circuitry die **206** and the sensor die **208** are stacked or vertically arranged in the z-direction in the image sensor device **210**. The interconnect layer **214** of the circuitry die **206** and the interconnect layer **218** of the sensor die **208** are facing toward each other in the image sensor die **210**, and the device layer **212** of the circuitry die **206** and the device layer **216** of the sensor die **208** are facing away from each other.

[0126] As shown in FIG. 8C, the DTI structure **252** may be formed in the substrate **246** (e.g., in the back side of the substrate **246**) and around the photodiodes **112** in the substrate **246**. For example, a deposition tool, an exposure tool, and a developer tool may be used to form a patterned masking layer on the substrate **246**. An etch tool may be used to etch the substrate **246** (e.g., using a wet etch technique, a dry etch technique) from the back side of the substrate **246** to form trenches in the back side of the substrate **246**. The trenches are above the STI structures **250** and alongside the photodiodes **112**.

[0127] A deposition tool may be used to conformally deposit a dielectric liner **256** of the DTI structure **252** in the trenches and on the back side surface of the substrate **246** using a CVD technique, an ALD technique, and/or another conformal deposition technique. A deposition tool may be used to deposit the dielectric material **254** of the DTI structure **252** on the dielectric liner **256** in the trenches and above the substrate **246** using a CVD technique, a PVD technique, an ALD technique, an oxidation technique, and/or another deposition technique. In some implementations, a planarization tool may perform a planarization operation to planarize the dielectric material **254** above the substrate **246**, which may remain as a buffer layer.

[0128] As shown in FIG. 8D, a passivation layer **266** may be formed over and/or on the buffer layer, and a metal layer **272** may be formed over and/or on the passivation layer over the back side of the substrate **246**. A deposition tool and/or a plating tool may be used to deposit the passivation layer **266** using a CVD technique, a PVD technique, an ALD technique, an oxidation technique, and/or another deposition technique. In some implementations, a planarization tool may perform a planarization operation to planarize the passivation layer **266** after the passivation layer **266** is deposited. A deposition tool and/or a plating tool may be used to deposit the metal layer **272** using a CVD technique, a PVD technique, an ALD technique, an electroplating technique, and/or another deposition technique. In some implementations, a planarization tool may perform a planarization operation to planarize the metal layer **272** after the metal layer **272** is deposited.

[0129] As shown in FIG. 8E, various layers and/or structures may be formed in the bonding pad region **228** of the image sensor device **210**. For example, a recess may be formed through the metal layer **272**, through the passivation layer **266**, through the buffer layer (corresponding to the dielectric material **254** above the substrate **246**), through the dielectric liner **256**, and/or through the substrate **246** to the dielectric layer **248**. In some implementations, a deposition tool, an exposure

tool, and a developer tool may be used to form a patterned masking layer on the metal layer **272**. An etch tool may be used to etch the through the metal layer **272**, through the passivation layer **266**, through the buffer layer, through the dielectric liner **256**, through the substrate **246** (e.g., using a wet etch technique, a dry etch technique) from the back side of the substrate **246** to form the recess.

[0130] A dielectric layer **274** may be formed in the recess on the dielectric layer **248**. A dielectric layer **276** may be formed on the dielectric layer **274**. A deposition tool may be used to deposit the dielectric layers **274** and **276** in the recess using a CVD technique, a PVD technique, an ALD technique, an oxidation technique, and/or another deposition technique.

[0131] Openings may be formed through the dielectric layers **248**, **274**, and **276** such that an interconnect structure **262** in the interconnect layer **218** is exposed through the recesses. A bonding pad structure **284** may be formed in the openings such that the bonding pad structure **284** lands on the interconnect structure **262**. The bonding pad structure **284** is also formed on the dielectric layer **276**.

[0132] In some implementations, a deposition tool, an exposure tool, and a developer tool may be used to form a patterned masking layer on the dielectric layer **276**. An etch tool may be used to etch the through the dielectric layer **276**, through the dielectric layer **274**, and through the dielectric layer **248** (e.g., using a wet etch technique, a dry etch technique) to form the recesses. A deposition tool and/or a plating tool may be used to deposit the bonding pad structure **284** in the recess using a CVD technique, a PVD technique, an ALD technique, an electroplating technique, and/or another deposition technique.

[0133] A dielectric layer **278** may be formed on the bonding pad structure **284**, and dielectric layers **280** and **282** may be deposited to fill in the recess in the bonding pad region **228**. A bonding pad opening **286** may be formed through the dielectric layers **278**, **280**, and **282** to expose the bonding pad structure **284**.

[0134] A deposition tool may be used to deposit the dielectric layers **278**, **280**, and **282** in the recess using a CVD technique, a PVD technique, an ALD technique, an oxidation technique, and/or another deposition technique. In some implementations, a deposition tool, an exposure tool, and a developer tool may be used to form a patterned masking layer on the dielectric layer **282**. An etch tool may be used to etch the through the dielectric layers **278**, **280**, and **282** (e.g., using a wet etch technique, a dry etch technique) to form the bonding pad opening **286**.

[0135] As shown in FIG. **8F**, the metal layer **272** in the pixel sensor array **222** is etched to form the metal grid structure **224**. Techniques described in connection with FIGS. **9A-9F**, **10A-10D**, **11A-11E**, **12A-12C**, **13A-13D**, **14A**, and/or **14B** may be used to form antireflection structures in and/or on the metal grid structure **224**.

[0136] In some implementations, a deposition tool, an exposure tool, and a developer tool may be used to form a patterned masking layer on the metal layer **272**. An etch tool may be used to etch the through the metal layer **272** to the passivation layer **266** (e.g., using a wet etch technique, a dry etch technique) remove portions of the metal layer **272**. Remaining portions of the metal layer **272** in the pixel sensor array **222** correspond to the metal grid structure **224** above the DTI structure **252**.

[0137] As further shown in FIG. **8F**, color filter regions **268** are formed in openings in the metal grid structure **224** such that the color filter regions **268** are located above and/or over the photodiodes **112** of the pixel sensors **100**. Micro-lenses **270** are formed on the color filter regions **268**.

[0138] As indicated above, FIGS. **8A-8F** are provided as an example. Other examples may differ from what is described with regard to FIGS. **8A-8F**.

[0139] FIGS. **9A-9F** are diagrams of an example implementation **900** of forming nanostructures **304** on a metal grid structure **224** of a pixel sensor array **222** described herein. The techniques described in connection with FIGS. **9A-9F** may be used to form nanostructures **304** in the examples of the pixel sensor arrays **222** illustrated and described in connection with FIGS. **3A** and **3C** (e.g.,

the examples in which the nanostructures **304** are included on the top surfaces **306** of the metal grid structures **224**). In some implementations, one or more of the semiconductor processing operations described in connection with FIGS. **9A-9F** may be performed using one or more semiconductor processing tools, such as a deposition tool, an exposure tool, a developer tool, an etch tool, a planarization tool, a plating tool, an ion implantation tool, and/or a wafer/die transport tool, among other examples.

[0140] As shown in FIG. **9A**, the metal grid structure **224** may be formed above the backside surface of the substrate **246** such that the metal grid structure **224** is included above the DTI structure **252**. An etch tool may be used to perform an etch operation to form the metal grid structure **224** from a metal layer (e.g., the metal layer **272**). In some implementations, a dry etch technique such as a plasma-based etch technique is used to form the metal grid structure **224** such that the sidewalls **310** of the metal grid structure **224** are angled relative to a top surface **306** of the metal grid structure **224**. This results in formation of an approximately trapezoidal cross-sectional profile for the metal grid structure **224**. One or more parameters of the etch operation may be selected to etch the metal layer to form the approximately trapezoidal cross-sectional profile for the metal grid structure **224**. The one or more parameters may include, for example, a bias voltage, a pressure, a temperature, and/or another parameter. Alternatively, the one or more parameters of the etch operation may be selected such that the dry etch results in formation of an approximately square cross-sectional profile for the metal grid structure **224**. The one or more parameters selected for the etch operation may result in a more vertical etch to form the approximately square cross-sectional profile than for forming the approximately trapezoidal cross-sectional profile.

[0141] As shown in FIG. **9B**, an antireflective layer **902** may be formed over and/or on the metal grid structure **224**, as well as on the backside surface of the substrate **246**. A deposition tool may be used to deposit the antireflective layer **902** using a CVD technique, a PVD technique, an ALD technique, an oxidation technique, and/or another deposition technique. In particular, a conformal deposition technique may be used to conformally deposit the antireflective layer **902** such that the antireflective layer **902** conforms to the profile of the metal grid structure **224**. The antireflective layer **902** may include one or more antireflective materials, such as zinc oxide (ZnO), aluminum-doped zinc oxide (Al: ZnO or AZO), and/or aluminum oxide (Al.sub.xO.sub.y such as Al.sub.3O.sub.4), among other examples. The antireflective layer **902** may be formed as a substrate on which the nanostructures **304** are formed or grown.

[0142] As shown in FIG. **9C**, a patterned masking layer **904** may be formed on the top surfaces **306** of the metal grid structure **224**. A deposition tool may be used to form a masking layer on the backside of the substrate **246** and on the metal grid structure **224** (e.g., using a spin-coating technique and/or another deposition technique). An exposure tool may be used to expose the masking layer to form a pattern in the masking layer. A developer tool may be used to develop the pattern to form the patterned masking layer **904**.

[0143] As shown in FIG. **9D**, an etch tool may be used to etch the antireflective layer **902** to remove portions of the antireflective layer **902** from the backside of the substrate **246** and from the sidewalls **310** of the metal grid structure **224**. Thus, the antireflective layer **902** remains only on the top surfaces **306** of the metal grid structure **224**. The portions of the antireflective layer **902** may be removed using a wet etch technique, a dry etch technique, and/or another suitable etch technique.

[0144] As shown in FIG. **9E**, the patterned masking layer **904** may be removed from the top surfaces **306** of the metal grid structure **224**. The patterned masking layer **904** may be removed by etching the patterned masking layer **904**, by ashing the patterned masking layer **904**, and/or using another suitable masking layer removal technique.

[0145] As shown in FIG. **9F**, the nanostructures **304** are formed on the antireflective layer **902** that is on the top surfaces **306** of the metal grid structure **224**. A thermal anneal operation may be performed to form the nanostructures **304**. For example, carbon (or graphite) powder may be deposited onto the antireflective layer **902** and annealed to cause the nanostructures **304** to grow

from the carbon powder in a highly crystalline manner.

[0146] As indicated above, FIGS. **9A-9F** are provided as an example. Other examples may differ from what is described with regard to FIGS. **9A-9F**.

[0147] FIGS. **10A-10D** are diagrams of an example implementation **1000** of forming nanostructures **304** on a metal grid structure **224** of a pixel sensor array **222** described herein. The techniques described in connection with FIGS. **10A-10D** may be used to form nanostructures **304** in the examples of the pixel sensor arrays **222** illustrated and described in connection with FIGS. **3B** and **3D** (e.g., the examples in which the nanostructures **304** are included on the top surfaces **306** and sidewalls **310** of the metal grid structures **224**). In some implementations, one or more of the semiconductor processing operations described in connection with FIGS. **10A-10D** may be performed using one or more semiconductor processing tools, such as a deposition tool, an exposure tool, a developer tool, an etch tool, a planarization tool, a plating tool, an ion implantation tool, and/or a wafer/die transport tool, among other examples.

[0148] The example implementation **1000** is similar to the example implementation **900** of forming nanostructures **304** in that the metal grid structure **224** is formed, the antireflective layer **902** is formed on the top surfaces **306** and the sidewalls **310** of the metal grid structure **224** as well as on the backside of the substrate **246**, and a patterned masking layer **904** is used to remove portions of the antireflective layer **902**.

[0149] However, and as shown in FIG. **10A**, the patterned masking layer **904** is also formed on the sidewalls **310** of the metal grid structure **224** in addition to the top surfaces **306** of the metal grid structure **224**. As shown in FIG. **10B**, this enables the portions of the antireflective layer **902** to be removed from the backside of the substrate **246** without removing the antireflective layer **902** from the top surfaces **306** and the sidewalls **310** of the metal grid structure **224**. Thus, in the example implementation **1000** of forming nanostructures **304**, the antireflective layer **902** is included on the top surfaces **306** and the sidewalls **310** of the metal grid structure **224**. As shown in FIG. **10C**, the patterned masking layer **904** is removed after the antireflective layer **902** is etched.

[0150] As shown in FIG. **10D**, retaining the antireflective layer **902** on the sidewalls **310** of the metal grid structure **224** in the example implementation **1000** of forming nanostructures **304** enables the nanostructures **304** to be grown on the sidewalls **310** of the metal grid structure **224** in addition to the top surfaces **306** of the metal grid structure **224**.

[0151] As indicated above, FIGS. **10A-10D** are provided as an example. Other examples may differ from what is described with regard to FIGS. **10A-10D**.

[0152] FIGS. **11A-11E** are diagrams of an example implementation **1100** of forming antireflective layers **402** on a metal grid structure **224** of a pixel sensor array **222** described herein. The techniques described in connection with FIGS. **11A-11E** may be used to form antireflective layers **402** in the examples of the pixel sensor arrays **222** illustrated and described in connection with FIGS. **4A** and **4C** (e.g., the examples in which the antireflective layers **402** are included on the top surfaces **306** of the metal grid structures **224**). In some implementations, one or more of the semiconductor processing operations described in connection with FIGS. **11A-11E** may be performed using one or more semiconductor processing tools, such as a deposition tool, an exposure tool, a developer tool, an etch tool, a planarization tool, a plating tool, an ion implantation tool, and/or a wafer/die transport tool, among other examples.

[0153] As shown in FIG. **11A**, the metal grid structure **224** may be formed above the backside surface of the substrate **246** such that the metal grid structure **224** is included above the DTI structure **252**. An etch tool may be used to perform an etch operation to form the metal grid structure **224** from a metal layer (e.g., the metal layer **272**). In some implementations, a dry etch technique such as a plasma-based etch technique is used to form the metal grid structure **224** such that the sidewalls **310** of the metal grid structure **224** are angled relative to a top surface **306** of the metal grid structure **224**. This results in formation of an approximately trapezoidal cross-sectional profile for the metal grid structure **224**. One or more parameters of the etch operation may be

selected to etch the metal layer to form the approximately trapezoidal cross-sectional profile for the metal grid structure **224**. The one or more parameters may include, for example, a bias voltage, a pressure, a temperature, and/or another parameter. Alternatively, the one or more parameters of the etch operation may be selected such that the dry etch results in formation of an approximately square cross-sectional profile for the metal grid structure **224**. The one or more parameters selected for the etch operation may result in a more vertical etch to form the approximately square cross-sectional profile than for forming the approximately trapezoidal cross-sectional profile.

[0154] As shown in FIG. **11B**, the antireflective layer **402** may be formed over and/or on the metal grid structure **224**, as well as on the backside surface of the substrate **246**. A deposition tool may be used to deposit the antireflective layer **902** using a CVD technique, a PVD technique, an ALD technique, an oxidation technique, and/or another deposition technique. In particular, a conformal deposition technique may be used to conformally deposit the antireflective layer **402** such that the antireflective layer **402** conforms to the profile of the metal grid structure **224**.

[0155] As shown in FIG. **11C**, a patterned masking layer **1102** may be formed on the top surfaces **306** of the metal grid structure **224**. A deposition tool may be used to form a masking layer on the backside of the substrate **246** and on the metal grid structure **224** (e.g., using a spin-coating technique and/or another deposition technique). An exposure tool may be used to expose the masking layer to form a pattern in the masking layer. A developer tool may be used to develop the pattern to form the patterned masking layer **1102**.

[0156] As shown in FIG. **11D**, an etch tool may be used to etch the antireflective layer **402** remove portions of the antireflective layer **402** from the backside of the substrate **246** and from the sidewalls **310** of the metal grid structure **224**. Thus, the antireflective layer **402** remains only on the top surfaces **306** of the metal grid structure **224**. The portions of the antireflective layer **402** may be removed using a wet etch technique, a dry etch technique, and/or another suitable etch technique.

[0157] As shown in FIG. **11E**, the patterned masking layer **1102** may be removed from the top surfaces **306** of the metal grid structure **224**. The patterned masking layer **1102** may be removed by etching the patterned masking layer **1102**, by ashing the patterned masking layer **1102**, and/or using another suitable masking layer removal technique.

[0158] As indicated above, FIGS. **11A-11E** are provided as an example. Other examples may differ from what is described with regard to FIGS. **11A-11E**.

[0159] FIGS. **12A-12C** are diagrams of an example implementation **1200** of forming antireflective layers **402** on a metal grid structure **224** of a pixel sensor array **222** described herein. The techniques described in connection with FIGS. **12A-12C** may be used to form antireflective layers **402** in the examples of the pixel sensor arrays **222** illustrated and described in connection with FIGS. **4B** and **4D** (e.g., the examples in which the antireflective layers **402** are included on the top surfaces **306** and sidewalls **310** of the metal grid structures **224**). In some implementations, one or more of the semiconductor processing operations described in connection with FIGS. **12A-12C** may be performed using one or more semiconductor processing tools, such as a deposition tool, an exposure tool, a developer tool, an etch tool, a planarization tool, a plating tool, an ion implantation tool, and/or a wafer/die transport tool, among other examples.

[0160] The example implementation **1200** is similar to the example implementation **1100** of forming antireflective layers **402** in that the metal grid structure **224** is formed, the antireflective layers **402** is formed on the top surfaces **306** and the sidewalls **310** of the metal grid structure **224** as well as on the backside of the substrate **246**, and a patterned masking layer **1102** is used to remove portions of the antireflective layer **902**.

[0161] However, and as shown in FIG. **12A**, the patterned masking layer **1102** is also formed on the sidewalls **310** of the metal grid structure **224** in addition to the top surfaces **306** of the metal grid structure **224**. As shown in FIG. **12B**, this enables the portions of the antireflective layer **402** to be removed from the backside of the substrate **246** without removing the antireflective layer **402** from the top surfaces **306** and the sidewalls **310** of the metal grid structure **224**. Thus, in the

example implementation **1200** of forming antireflective layers **402**, the antireflective layers **402** are included on the top surfaces **306** and on the sidewalls **310** of the metal grid structure **224**. As shown in FIG. **12C**, the patterned masking layer **904** is removed after the antireflective layer **402** is etched. [0162] As indicated above, FIGS. **12A-12C** are provided as an example. Other examples may differ from what is described with regard to FIGS. **12A-12C**.

[0163] FIGS. **13A-13D** are diagrams of an example implementation **1300** of forming cavities **502** in a metal grid structure **224** of a pixel sensor array **222** described herein. The techniques described in connection with FIGS. **13A-13D** may be used to form cavities **502** in the examples of the pixel sensor arrays **222** illustrated and described in connection with FIGS. **5A, 5B, 5F, and 5G** (e.g., the examples in which the cavities **502** are included in the top surfaces **306** of the metal grid structures **224**). In some implementations, one or more of the semiconductor processing operations described in connection with FIGS. **13A-13D** may be performed using one or more semiconductor processing tools, such as a deposition tool, an exposure tool, a developer tool, an etch tool, a planarization tool, a plating tool, an ion implantation tool, and/or a wafer/die transport tool, among other examples.

[0164] As shown in FIG. **13A**, the metal grid structure **224** may be formed above the backside surface of the substrate **246** such that the metal grid structure **224** is included above the DTI structure **252**. An etch tool may be used to perform an etch operation to form the metal grid structure **224** from a metal layer (e.g., the metal layer **272**). In some implementations, a dry etch technique such as a plasma-based etch technique is used to form the metal grid structure **224** such that the sidewalls **310** of the metal grid structure **224** are angled relative to a top surface **306** of the metal grid structure **224**. This results in formation of an approximately trapezoidal cross-sectional profile for the metal grid structure **224**. One or more parameters of the etch operation may be selected to etch the metal layer to form the approximately trapezoidal cross-sectional profile for the metal grid structure **224**. The one or more parameters may include, for example, a bias voltage, a pressure, a temperature, and/or another parameter. Alternatively, the one or more parameters of the etch operation may be selected such that the dry etch results in formation of an approximately square cross-sectional profile for the metal grid structure **224**. The one or more parameters selected for the etch operation may result in a more vertical etch to form the approximately square cross-sectional profile than for forming the approximately trapezoidal cross-sectional profile.

[0165] As shown in FIGS. **13B and 13C**, a masking layer **1302** may be formed on the metal grid structure **224** and on the backside of the substrate **246**, and a pattern **1304** may be formed in the masking layer **1302**. A deposition tool may be used to form a masking layer **1302** on the backside of the substrate **246** and on the metal grid structure **224** (e.g., using a spin-coating technique and/or another deposition technique). An exposure tool may be used to expose the masking layer **1302** to form the pattern **1304** in the masking layer. A developer tool may be used to develop the pattern **1304**. The pattern **1304** may include openings in the masking layer **1302** above the top surfaces **306** of the metal grid structure **224**.

[0166] An etch tool may be used to etch the top surfaces **306** of the metal grid structure **224** based on the pattern **1304** in the masking layer **1302** to form the cavities **502** in the top surfaces **306** of the metal grid structure **224**, as shown in FIG. **13D**.

[0167] In some implementations, the cavities **502** may be formed using a wet etch technique, in which a wet etchant is provided to the top surfaces **306** of the metal grid structure **224** through the pattern **1304** in the masking layer **1302**, and the wet etchant removes material from the top surfaces **306** of the metal grid structure **224**. This results in formation of the cavities in the top surfaces **306** of the metal grid structure **224**. In some implementations, using a wet etch technique results in the cavities **502** having approximately curved, rounded, and/or semi-circular cross-sectional profiles.

[0168] In some implementations, the cavities **502** may be formed using a dry etch technique, in which a plasma and/or another type of dry etchant is provided to the top surfaces **306** of the metal grid structure **224** through the pattern **1304** in the masking layer **1302**, and the dry etchant removes

material from the top surfaces **306** of the metal grid structure **224**. This results in formation of the cavities in the top surfaces **306** of the metal grid structure **224**. In some implementations, using a dry etch technique results in the cavities **502** having approximately V-shaped (or triangular) cross-sectional profiles.

[0169] As indicated above, FIGS. **13A-13D** are provided as an example. Other examples may differ from what is described with regard to FIGS. **13A-13D**.

[0170] FIGS. **14A** and **14B** are diagrams of an example implementation **1400** of forming cavities **502** in a metal grid structure **224** of a pixel sensor array **222** described herein. The techniques described in connection with FIGS. **14A** and **14B** may be used to form cavities **502** in the examples of the pixel sensor arrays **222** illustrated and described in connection with FIGS. **5C**, **5D**, **5E**, **5H**, **5I**, and **5J** (e.g., the examples in which the cavities **502** are included in the top surfaces **306** and in the sidewalls **310** of the metal grid structures **224**). In some implementations, one or more of the semiconductor processing operations described in connection with FIGS. **14A** and **14B** may be performed using one or more semiconductor processing tools, such as a deposition tool, an exposure tool, a developer tool, an etch tool, a planarization tool, a plating tool, an ion implantation tool, and/or a wafer/die transport tool, among other examples.

[0171] The example implementation **1400** is similar to the example implementation **1300** of forming cavities **502** in the metal grid structure **224**. However, as shown in FIG. **14A**, a pattern **1402** is formed in the masking layer **1302**, and the pattern **1402** includes openings over the sidewalls **310** of the metal grid structure **224** as well as over the top surfaces **306** of the metal grid structure **224**. An etch tool may be used to etch the top surfaces **306** of the metal grid structure **224** based on the pattern **1402** in the masking layer **1302** to form the cavities **502** in the top surfaces **306** of the metal grid structure **224** as well as to form cavities **502** in the sidewalls **310** of the metal grid structure **224**, as shown in FIG. **14B**.

[0172] As indicated above, FIGS. **14A** and **14B** are provided as an example. Other examples may differ from what is described with regard to FIGS. **14A** and **14B**.

[0173] FIG. **15** is a flowchart of an example process **1500** associated with forming a metal grid structure of a pixel sensor array. In some implementations, one or more process blocks of FIG. **15** are performed using one or more semiconductor processing tools, such as a deposition tool, an exposure tool, a developer tool, an etch tool, a planarization tool, a plating tool, an ion implantation tool, and/or a wafer/die transport tool, among other examples.

[0174] As shown in FIG. **15**, process **1500** may include forming a plurality of photodiodes in a substrate of a pixel sensor array (block **1510**). For example, one or more semiconductor processing tools may be used to form a plurality of photodiodes **112** in a substrate **246** of a pixel sensor array **222**, as described herein.

[0175] As further shown in FIG. **15**, process **1500** may include forming a DTI structure around the plurality of photodiodes in the substrate (block **1520**). For example, one or more semiconductor processing tools may be used to form a DTI structure **252** around the plurality of photodiodes **112** in the substrate **246**, as described herein.

[0176] As further shown in FIG. **15**, process **1500** may include forming a metal grid structure above the substrate and over the DTI structure (block **1530**). For example, one or more semiconductor processing tools may be used to form a metal grid structure **224** above the substrate **246** and over the DTI structure **252**, as described herein. In some implementations, the DTI structure **252** laterally surrounds the plurality of photodiodes **112**.

[0177] As further shown in FIG. **15**, process **1500** may include forming an antireflection structure at least one of in a top surface of the metal grid structure or on the top surface of the metal grid structure (block **1540**). For example, one or more semiconductor processing tools may be used to form an antireflection structure at least one of in a top surface **306** of the metal grid structure **224** or on the top surface **306** of the metal grid structure **224**, as described herein. In some implementations, the antireflection structure includes an antireflective layer **404** on the top surface

306 of the metal grid structure **224**. In some implementations, the antireflection structure includes a plurality of nanostructures **304** on the top surface **306** of the metal grid structure **224**. In some implementations, the antireflection structure includes a plurality of nanostructures **304** formed on an antireflective layer **902** on the top surface **306** of the metal grid structure **224**. In some implementations, the antireflection structure includes a plurality of cavities **502** formed in the top surface **306** of the metal grid structure **224**.

[0178] Process **1500** may include additional implementations, such as any single implementation or any combination of implementations described below and/or in connection with one or more other processes described elsewhere herein.

[0179] In a first implementation, forming the antireflection structure includes depositing a layer of dielectric material (e.g., the antireflective layer **402**, the antireflective layer **902**) on the metal grid structure **224** and on the semiconductor layer **246**, and removing portions of the layer of dielectric material such that remaining portions of the layer of dielectric material correspond to the antireflection layer (e.g., the antireflective layer **402**, the antireflective layer **902**) on the top surface **306** of the metal grid structure **224**.

[0180] In a second implementation, alone or in combination with the first implementation, removing the portions of the layer of dielectric material includes removing the portions of the layer of dielectric material from sidewalls **310** of the metal grid structure **224**.

[0181] In a third implementation, alone or in combination with one or more of the first and second implementations, process **1500** includes performing a thermal anneal operation to form nanowires (e.g., nanostructures **304**) on the antireflection layer.

[0182] In a fourth implementation, alone or in combination with one or more of the first through third implementations, forming the antireflection layer includes forming a pattern (e.g., a pattern **1304**, a pattern **1402**) in a masking layer **1302** on the metal grid structure **224**, and etching the top surface **306** of the metal grid structure **224** based on the pattern to form a plurality of cavities **502** in the top surface **306** of the metal grid structure **224**, wherein the plurality of cavities **502** correspond to the antireflection structure in the top surface **306** of the metal grid structure **224**.

[0183] Although FIG. **15** shows example blocks of process **1500**, in some implementations, process **1500** includes additional blocks, fewer blocks, different blocks, or differently arranged blocks than those depicted in FIG. **15**. Additionally, or alternatively, two or more of the blocks of process **1500** may be performed in parallel.

[0184] In this way, an image sensor device may include one or more types of antireflection structures on a metal grid surrounding the pixel sensors in a pixel sensor array of the image sensor device. The antireflection structures may include an antireflective layer, nanostructures extending from the antireflective layer, and/or a plurality of cavities formed in the metal grid structure. The antireflection structures may be included in and/or on one or more surfaces of the metal grid structure to reduce the reflection of incident light, which may reduce the likelihood and/or magnitude of optical crosstalk between adjacent pixel sensors in the pixel sensor array.

[0185] As described in greater detail above, some implementations described herein provide a pixel sensor array. The pixel sensor array includes a plurality of pixel sensors, arranged in a grid, comprising a plurality of photodiodes in a substrate. The pixel sensor array includes a DTI structure, laterally surrounding the plurality of photodiodes, in the substrate. The pixel sensor array includes a metal grid structure above the DTI structure and above the substrate, where the metal grid structure surrounds the photodiodes. The pixel sensor array includes an antireflective layer on a top surface of the metal grid structure. The top surface faces away from the substrate.

[0186] As described in greater detail above, some implementations described herein provide an image sensor device. The image sensor device includes a plurality of pixel sensors, arranged in a pixel sensor array, that include a plurality of photodiodes in a substrate of the image sensor device. The image sensor device includes a DTI structure, around the plurality of photodiodes, in the substrate. The image sensor device includes a metal grid structure above the DTI structure and

above the semiconductor layer, where the metal grid structure surrounds the photodiodes, and where the metal grid structure includes a plurality of concave surfaces in a top surface of the metal grid structure.

[0187] As described in greater detail above, some implementations described herein provide a method. The method includes forming a plurality of photodiodes in a substrate of a pixel sensor array. The method includes forming a DTI structure around the plurality of photodiodes in the substrate. The method includes forming a metal grid structure above the substrate and over the DTI structure. The method includes forming an antireflection structure at least one of in a top surface of the metal grid structure or on the top surface of the metal grid structure.

[0188] As used herein, “satisfying a threshold” may, depending on the context, refer to a value being greater than the threshold, greater than or equal to the threshold, less than the threshold, less than or equal to the threshold, equal to the threshold, not equal to the threshold, or the like.

[0189] The terms “approximately” and “substantially” can indicate a value of a given quantity that varies within 5% of the value (e.g., +1%, +2%, +3%, +4%, +5% of the value). These values are merely examples and are not intended to be limiting. It is to be understood that the terms “approximately” and “substantially” can refer to a percentage of the values of a given quantity in light of this disclosure.

[0190] The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A pixel sensor array, comprising: a plurality of pixel sensors, arranged in a grid, comprising a plurality of photodiodes in a substrate; a deep trench isolation (DTI) structure, laterally surrounding the plurality of photodiodes, in the substrate; a metal grid structure above the DTI structure and above the substrate, wherein the metal grid structure surrounds the photodiodes; and an antireflective layer on a top surface of the metal grid structure, wherein the top surface faces away from the substrate.
2. The pixel sensor array of claim 1, wherein the metal grid structure has a substantially trapezoidal cross-sectional profile.
3. The pixel sensor array of claim 1, wherein antireflective layer is further included on sidewalls of the metal grid structure.
4. The pixel sensor array of claim 1, wherein the metal grid structure has a substantially square-shaped cross-sectional profile.
5. The pixel sensor array of claim 1, wherein the metal grid structure further includes a plurality of nanowires on the antireflective layer.
6. The pixel sensor array of claim 5, wherein the plurality of nanowires comprises a plurality of carbon nanowires.
7. The pixel sensor array of claim 1, wherein the antireflective layer comprises at least one of: zinc oxide (ZnO.sub.x), aluminum oxide (Al.sub.xO.sub.y), or aluminum-doped zinc oxide (AZO).
8. An image sensor device comprising: a plurality of pixel sensors, arranged in a pixel sensor array, comprising a plurality of photodiodes in a substrate of the image sensor device; a deep trench isolation (DTI) structure, around the plurality of photodiodes, in the substrate; and a metal grid structure above the DTI structure and above the substrate, wherein the metal grid structure

surrounds the photodiodes, and wherein the metal grid structure comprises a plurality of concave surfaces in a top surface of the metal grid structure.

9. The image sensor device of claim 8, wherein the plurality of concave surfaces have a rounded cross-sectional profile.

10. The image sensor device of claim 8, wherein the metal grid structure has a substantially trapezoidal cross-sectional profile.

11. The image sensor device of claim 8, wherein the plurality of concave surfaces have a substantially V-shaped cross-sectional profile.

12. The image sensor device of claim 8, wherein the metal grid structure comprises another plurality of concave surfaces in sidewalls of the metal grid structure.

13. The image sensor device of claim 12, wherein a cross-sectional profile of the plurality of concave surfaces in the top surface of the metal grid structure, and a cross-sectional profile of the other plurality of concave surfaces in the sidewalls of the metal grid structure, are different cross-sectional profiles.

14. The image sensor device of claim 12, wherein a cross-sectional profile of the plurality of concave surfaces in the top surface of the metal grid structure, and a cross-sectional profile of the other plurality of concave surfaces in the sidewalls of the metal grid structure, are approximately a same cross-sectional profile.

15. The image sensor device of claim 8, wherein the metal grid structure has a substantially square-shaped cross-sectional profile.

16. A method, comprising: forming a plurality of photodiodes in a substrate of a pixel sensor array; forming a deep trench isolation (DTI) structure around the plurality of photodiodes in the substrate; forming a metal grid structure above the substrate and over the DTI structure; and forming an antireflection structure at least one of in a top surface of the metal grid structure or on the top surface of the metal grid structure.

17. The method of claim 16, wherein forming the antireflection structure comprises: depositing a layer of dielectric material on the metal grid structure and on the substrate; and removing portions of the layer of dielectric material such that remaining portions of the layer of dielectric material correspond to an antireflection layer on the top surface of the metal grid structure.

18. The method of claim 17, wherein removing the portions of the layer of dielectric material comprises: removing the portions of the layer of dielectric material from sidewalls of the metal grid structure.

19. The method of claim 17, further comprising: performing a thermal anneal operation to form nanowires on the antireflection layer.

20. The method of claim 16, wherein forming the antireflection structure comprises: forming a pattern in a masking layer on the metal grid structure; and etching the top surface of the metal grid structure based on the pattern to form a plurality of cavities in the top surface of the metal grid structure, wherein the plurality of cavities correspond to the antireflection structure in the top surface of the metal grid structure.
